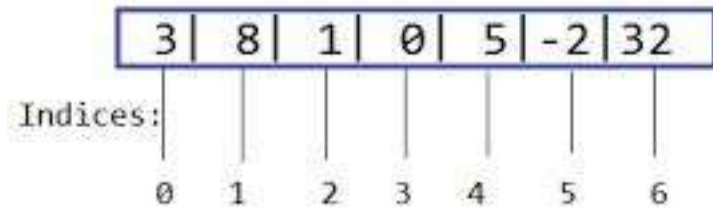


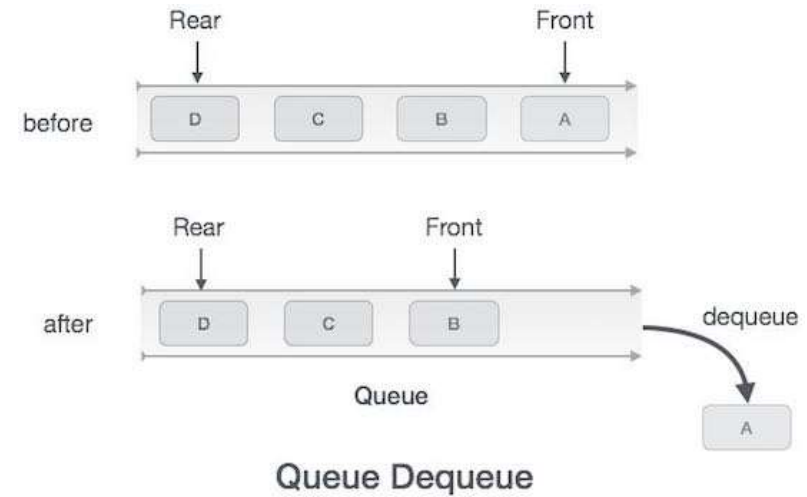
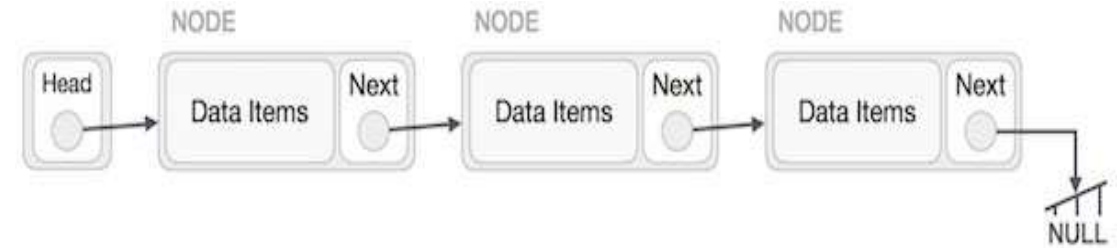
So far we discussed Linear data structures like

Array :



e
d
c
b
a

stack



Introduction to trees

- So far we have discussed mainly linear data structures – strings, arrays, lists, stacks and queues
- Now we will discuss a non-linear data structure called **tree**.
- Trees are mainly used to represent data containing a hierarchical relationship between elements, for example, records, family trees and table of contents.
- Consider a parent-child relationship

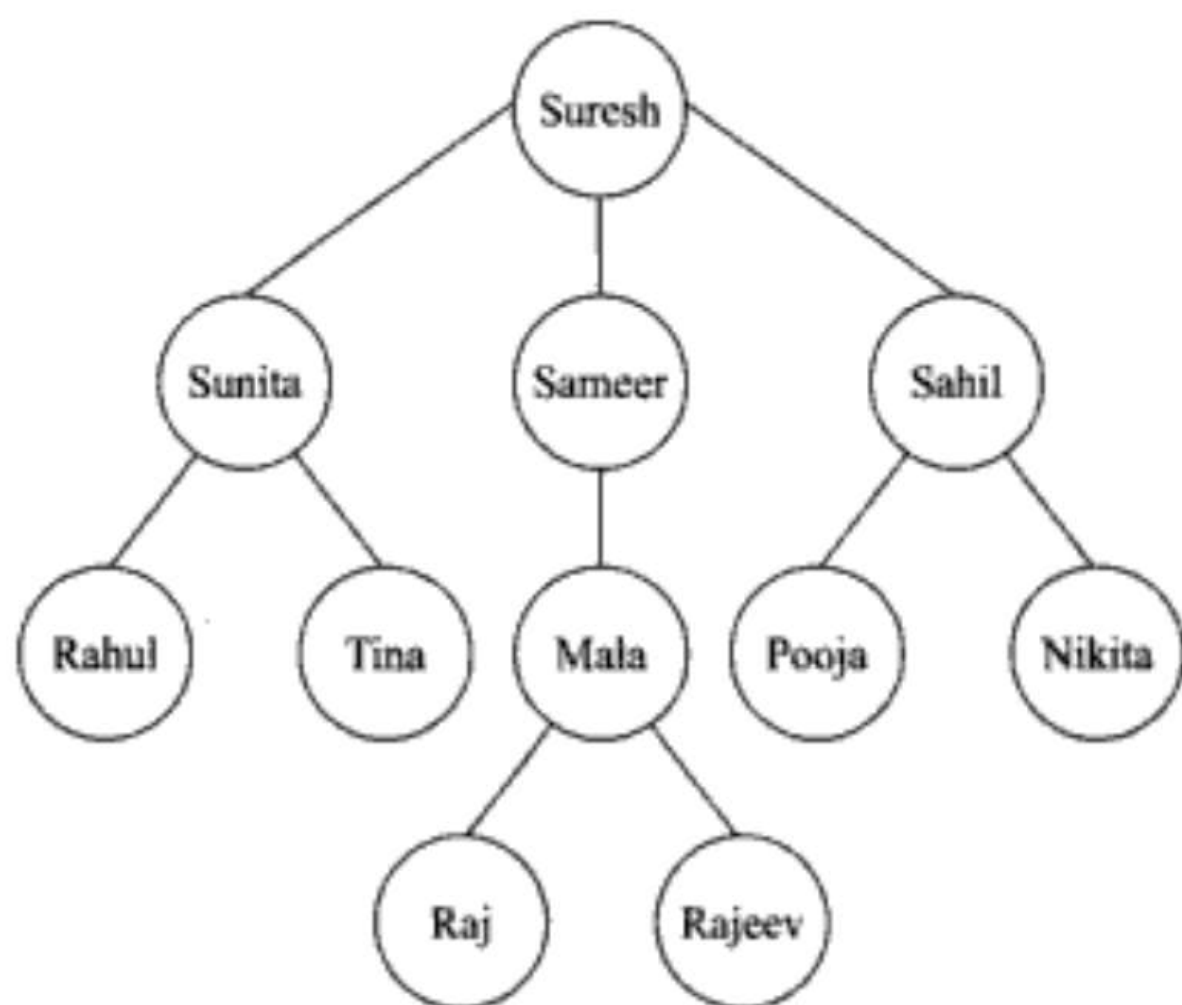
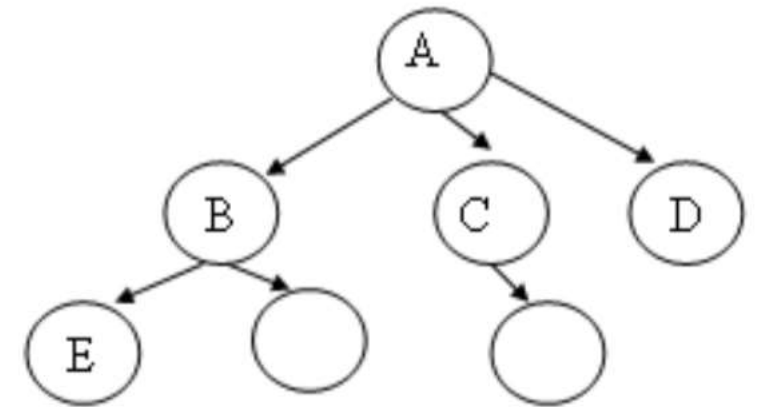


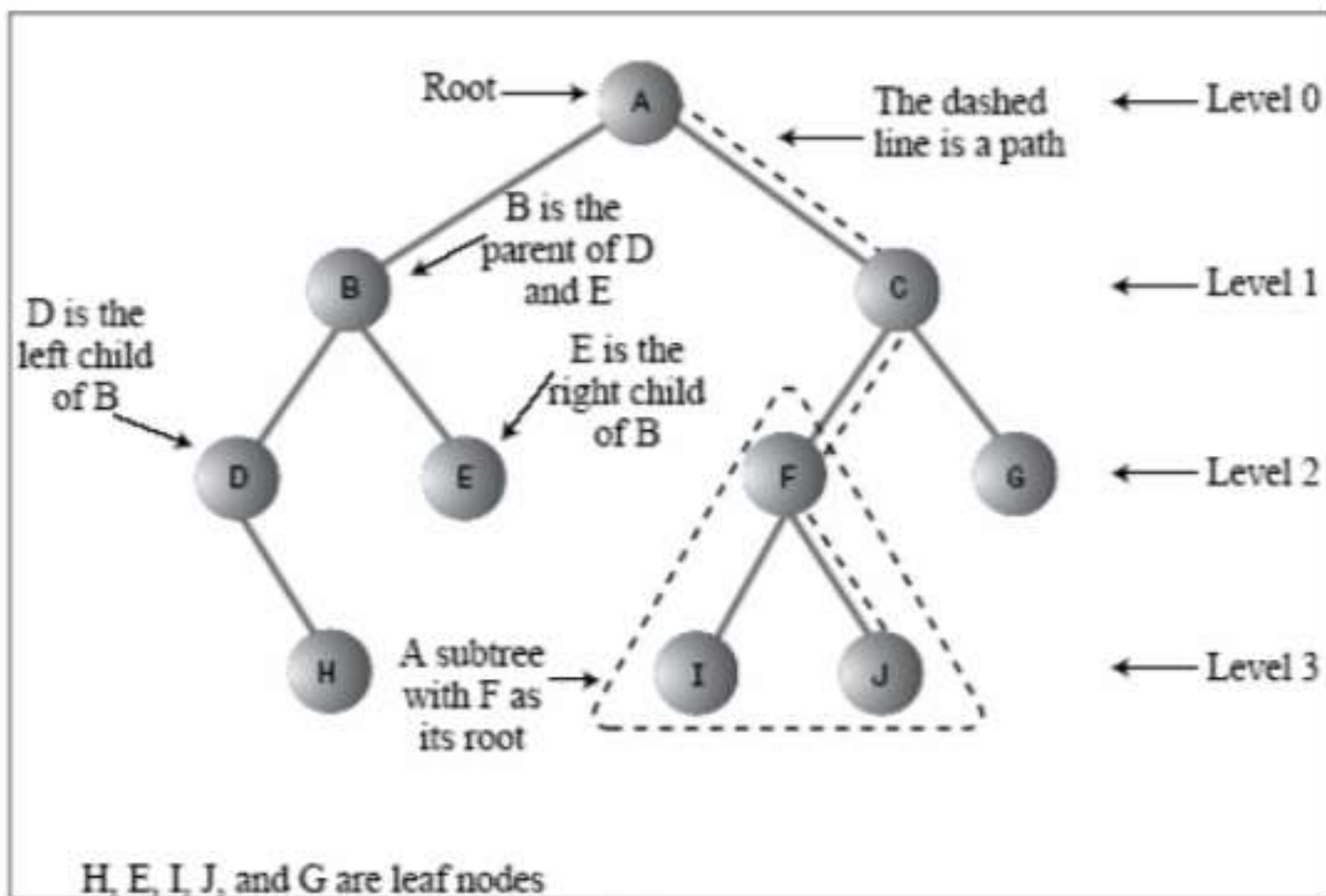
Fig. 8.1 *A Hypothetical Family Tree*

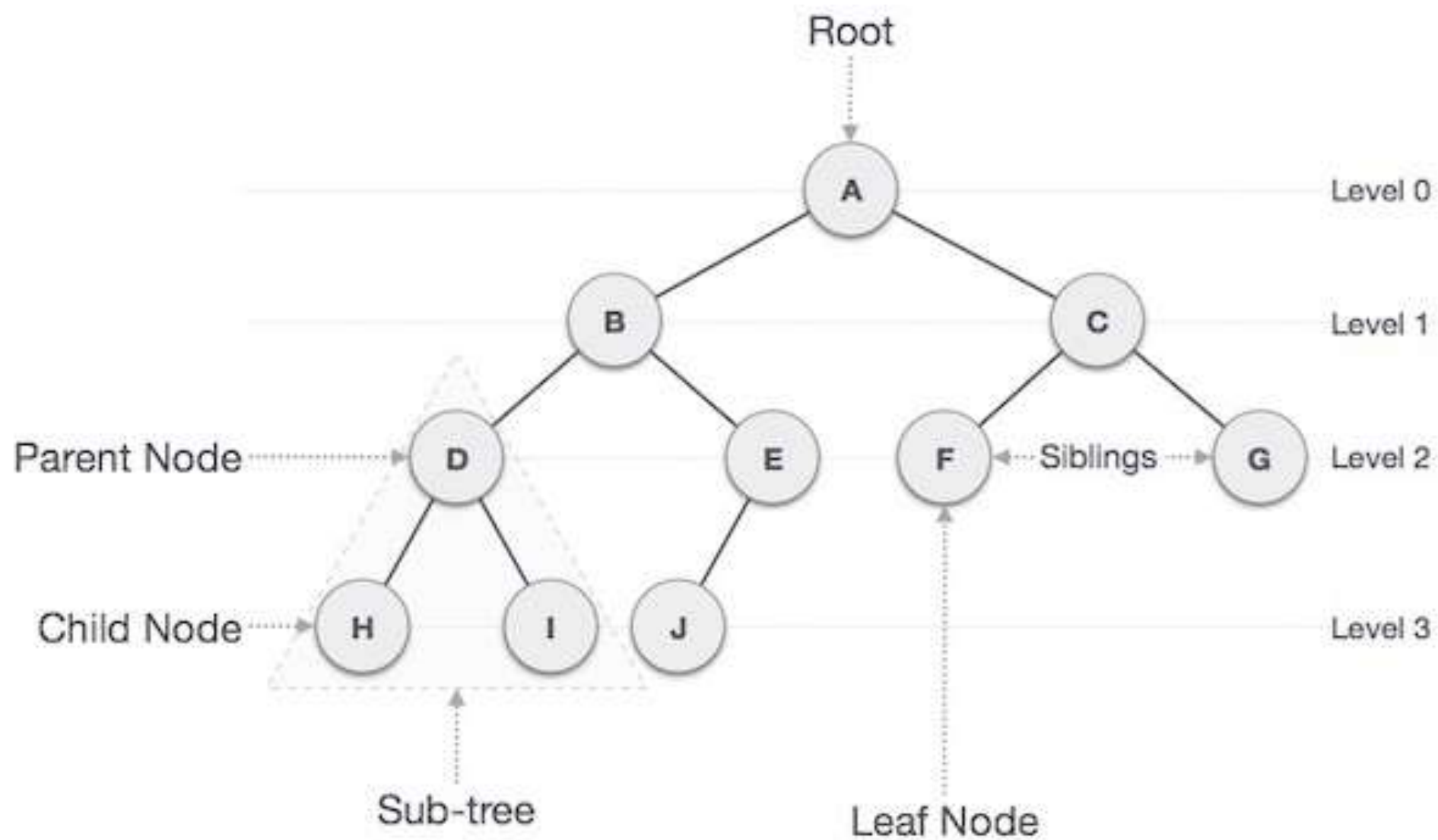
Tree

- A tree is an abstract model of a hierarchical structure that consists of nodes with a parent-child relationship.
 - Tree is a sequence of **nodes**
 - There is a starting node known as a **root** node
 - Every node other than the root has a **parent** node.
 - Nodes may have any number of children



A has 3 children, B, C, D
A is parent of B



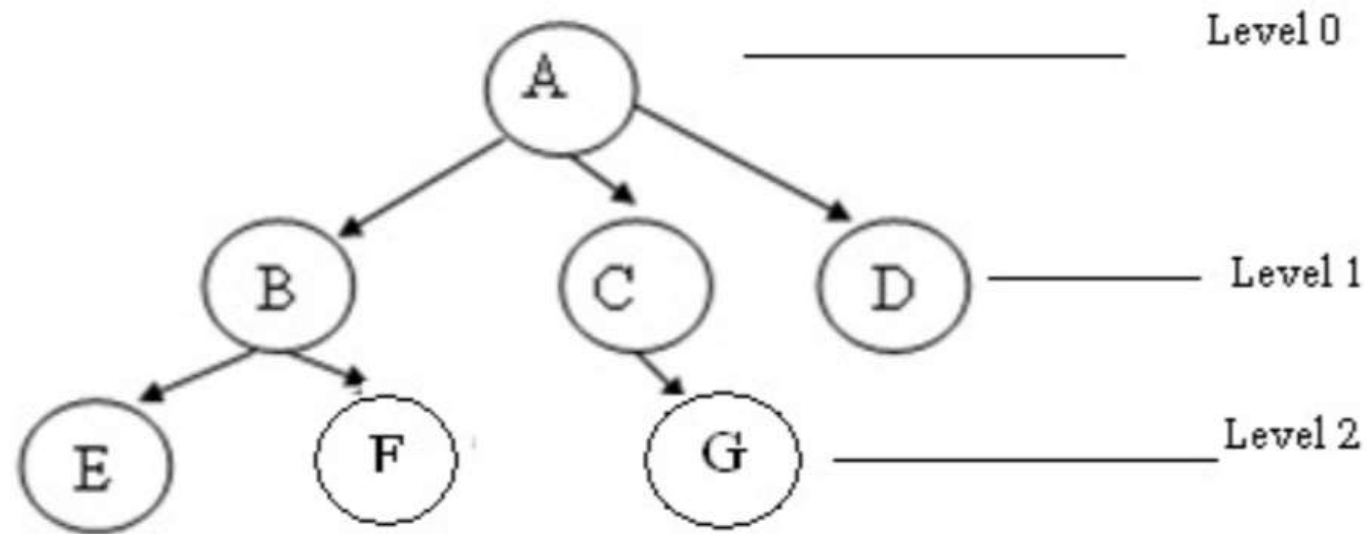


Some Key Terms:

- **Root** – Node at the top of the tree is called root.
- **Parent** – Any node except root node has one edge upward to a node called parent.
- **Child** – Node below a given node connected by its edge downward is called its child node.
- **Sibling** – Child of same node are called siblings
- **Leaf** – Node which does not have any child node is called leaf node.
- **Sub tree** – Sub tree represents descendants of a node.
- **Levels** – Level of a node represents the generation of a node. If root node is at level 0, then its next child node is at level 1, its grandchild is at level 2 and so on.
- **keys** – Key represents a value of a node based on which a search operation is to be carried out for a node.

Some Key Terms:

- Degree of a node:
 - The degree of a node is the number of children of that node
- Degree of a Tree:
 - The degree of a tree is the maximum degree of nodes in a given tree
- Path:
 - It is the sequence of consecutive edges from source node to destination node.
- Height of a node:
 - The height of a node is the max path length from that node to a leaf node.
- Height of a tree:
 - The height of a tree is the height of the root
- Depth of a tree:
 - Depth of a tree is the max level of any leaf in the tree



- ✓ A is the root node
- ✓ B is the parent of E and F
- ✓ D is the sibling of B and C
- ✓ E and F are children of B
- ✓ E, F, G, D are external nodes or leaves
- ✓ A, B, C are internal nodes
- ✓ Depth of F is 2
- ✓ the height of tree is 2
- ✓ the degree of node A is 3
- ✓ The degree of tree is 3

Characteristics of trees

- Non-linear data structure
- Combines advantages of an ordered array
- Searching as fast as in ordered array
- Insertion and deletion as fast as in linked list
- Simple and fast

Application

- Directory structure of a file store
- Structure of an arithmetic expressions
- Used in almost every 3D video game to determine what objects need to be rendered.
- Used in almost every high-bandwidth router for storing router-tables.
- used in compression algorithms, such as those used by the .jpeg and .mp3 file-formats.

Introduction To Binary Trees

- A **binary tree**, is a tree in which no node can have more than two children.
- Consider a binary tree T, here 'A' is the root node of the binary tree T.
- 'B' is the left child of 'A' and 'C' is the right child of 'A'
 - i.e A is a father of B and C.
 - The node B and C are called siblings.
- Nodes D,H,I,F,J are leaf node

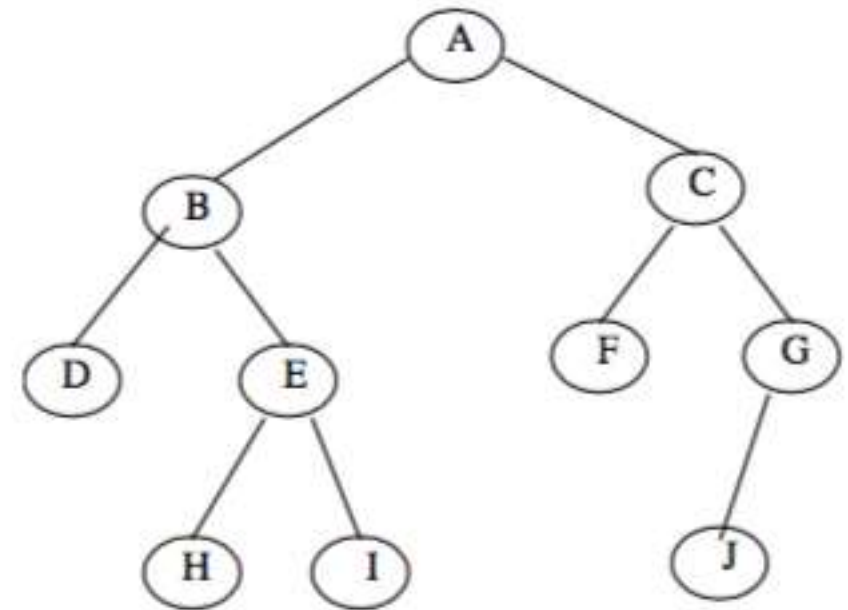
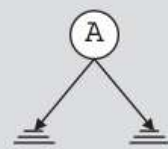


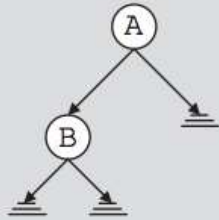
Fig. 8.3. Binary tree

Binary Trees

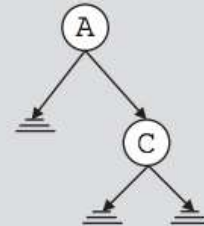
- A **binary tree**, T , is either empty or such that
 - T has a special node called the **root** node
 - T has two sets of nodes L_T and R_T , called the **left subtree** and **right subtree** of T , respectively.
 - L_T and R_T are binary trees.



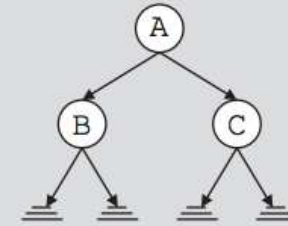
(a) Binary tree
with one node



(b) Binary tree
with two nodes



(c) Binary tree
with two nodes

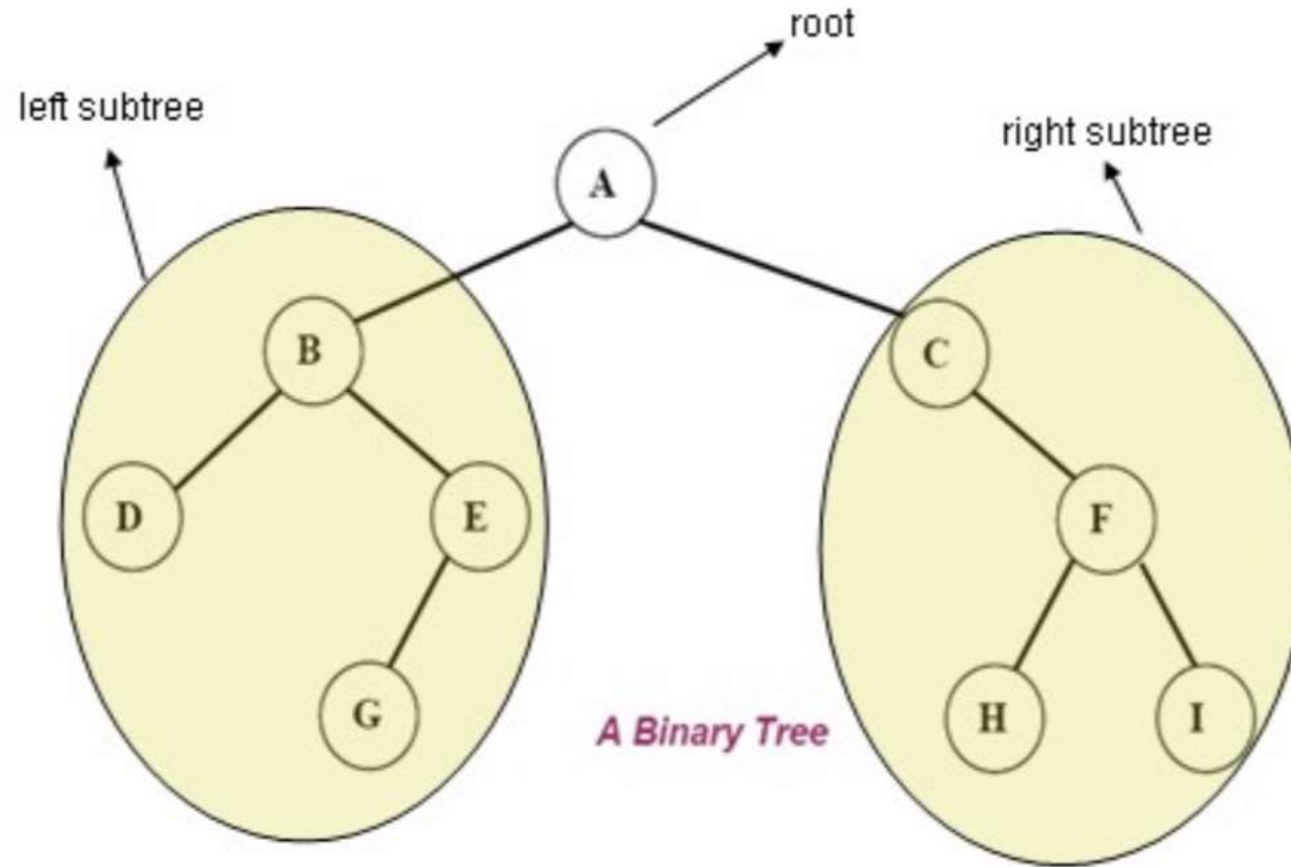


(d) Binary tree
with three nodes

Binary Tree

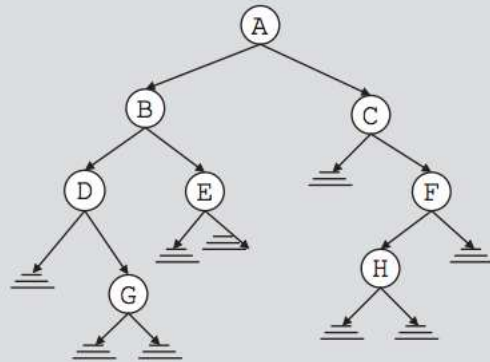
- A binary tree is a finite set of elements that are either empty or is partitioned into three disjoint subsets.
- The first subset contains a single element called the **root** of the tree.
- The other two subsets are themselves binary trees called the **left** and **right sub-trees** of the original tree.
- A left or right sub-tree can be empty.
- Each element of a binary tree is called a **node** of the tree.

The following figure shows a binary tree with 9 nodes where A is the root



Binary Tree

- The root node of this binary tree is A.
- The left sub tree of the root node, which we denoted by L_A , is the set $L_A = \{B, D, E, G\}$ and the right sub tree of the root node, R_A is the set $R_A = \{C, F, H\}$
- The root node of L_A is node B, the root node of R_A is C and so on



Binary Tree Properties

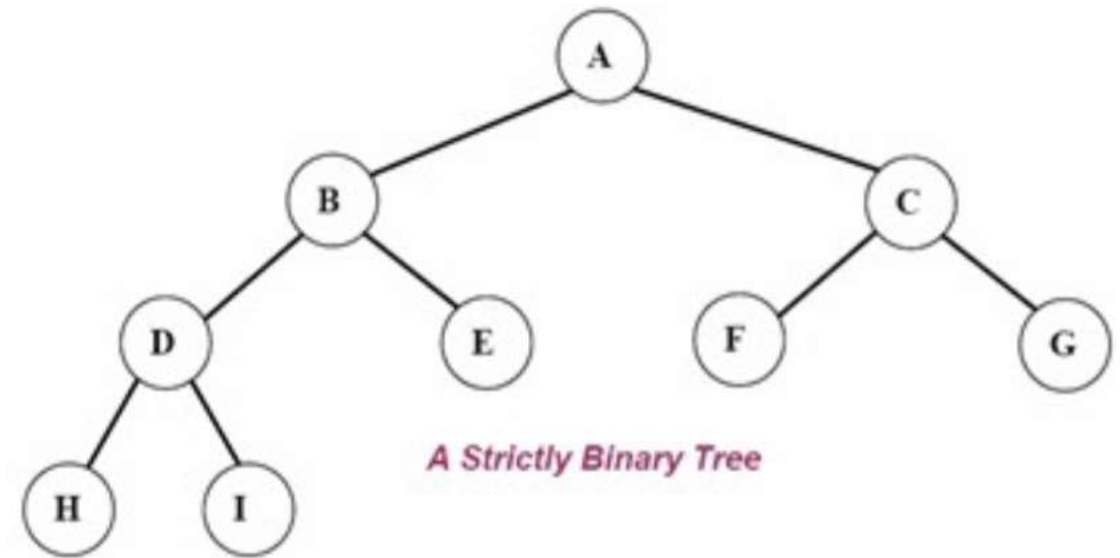
- If a binary tree contains m nodes at level L , it contains at most $2m$ nodes at level $L+1$
- Since a binary tree can contain at most 1 node at level 0 (the root), it contains at most 2^L nodes at level L .

Types of Binary Tree

- Complete binary tree
- Strictly binary tree
- Almost complete binary tree

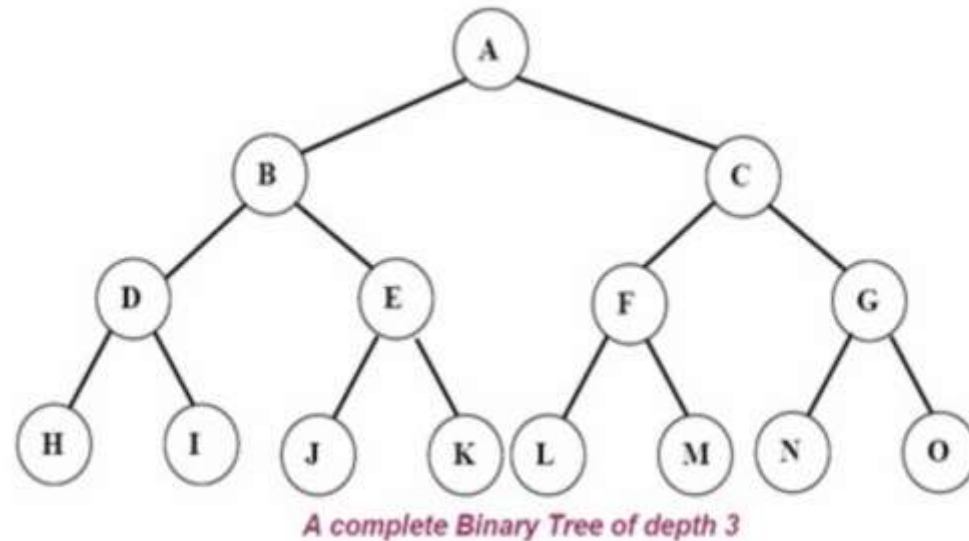
Strictly binary tree

- If every non-leaf node in a binary tree has nonempty left and right sub-trees, then such a tree is called a **strictly binary tree**.
- **Or, to put it another way**, all of the nodes in a strictly binary tree are of degree zero or two, never degree one.
- A strictly binary tree with N leaves always contains $2N - 1$ nodes.



Complete binary tree

- A **complete binary tree** is a **binary tree** in which every level, except possibly the last, is completely filled, and all nodes are as far left as possible.
- A *complete binary tree* of depth **d** is called strictly binary tree if all of whose leaves are at level **d**.
- A complete binary tree has 2^d nodes at every depth **d** and $2^d - 1$ non leaf nodes



Almost complete binary tree

- An almost complete binary tree is a tree where for a right child, there is always a left child, but for a left child there may not be a right child.

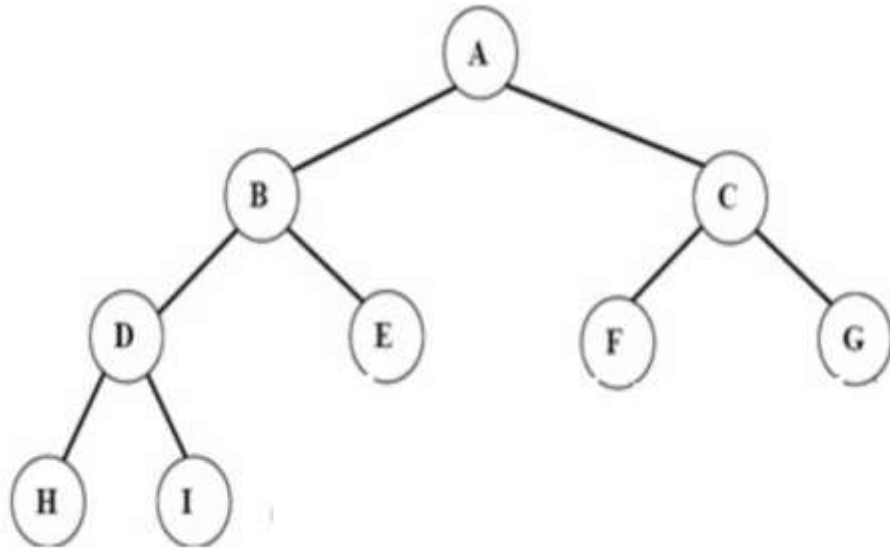


Fig Almost complete binary tree.

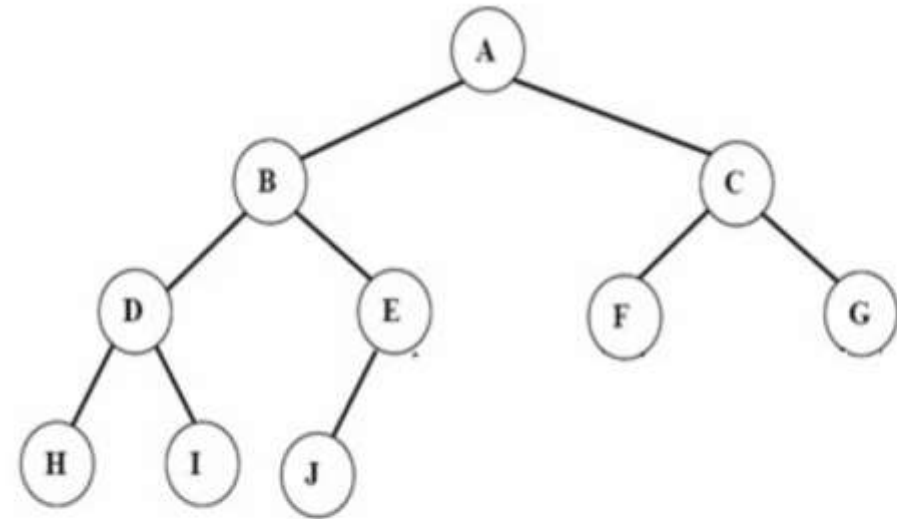


Fig Almost complete binary tree but not strictly binary tree.
Since node E has a left son but not a right son.

Operations on Binary tree:

- ✓ **father(n,T):**Return the parent node of the node n in tree T. If n is the root, NULL is returned.
- ✓ **LeftChild(n,T):**Return the left child of node n in tree T. Return NULL if n does not have a left child.
- ✓ **RightChild(n,T):**Return the right child of node n in tree T. Return NULL if n does not have a right child.
- ✓ **Info(n,T):** Return information stored in node n of tree T (ie. Content of a node).
- ✓ **Sibling(n,T):** return the sibling node of node n in tree T. Return NULL if n has no sibling.
- ✓ **Root(T):** Return root node of a tree if and only if the tree is nonempty.
- ✓ **Size(T):** Return the number of nodes in tree T
- ✓ **MakeEmpty(T):** Create an empty tree T
- ✓ **SetLeft(S,T):** Attach the tree S as the left sub-tree of tree T
- ✓ **SetRight(S,T):** Attach the tree S as the right sub-tree of tree T.
- ✓ **Preorder(T):** Traverses all the nodes of tree T in preorder.
- ✓ **postorder(T):** Traverses all the nodes of tree T in postorder
- ✓ **Inorder(T):** Traverses all the nodes of tree T in inorder.

Tree traversal

- Traversal is a process to visit all the nodes of a tree and may print their values too.
- All nodes are connected via edges (links) we always start from the root (head) node.
- There are three ways which we use to traverse a tree
 - In-order Traversal
 - Pre-order Traversal
 - Post-order Traversal
- Generally we traverse a tree to search or locate given item or key in the tree or to print all the values it contains.

Pre-order, In-order, Post-order

- Pre-order

<root><left><right>

- In-order

<left><root><right>

- Post-order

<left><right><root>

Pre-order Traversal

- The preorder traversal of a nonempty binary tree is defined as follows:
 - Visit the root node
 - Traverse the left sub-tree in preorder
 - Traverse the right sub-tree in preorder

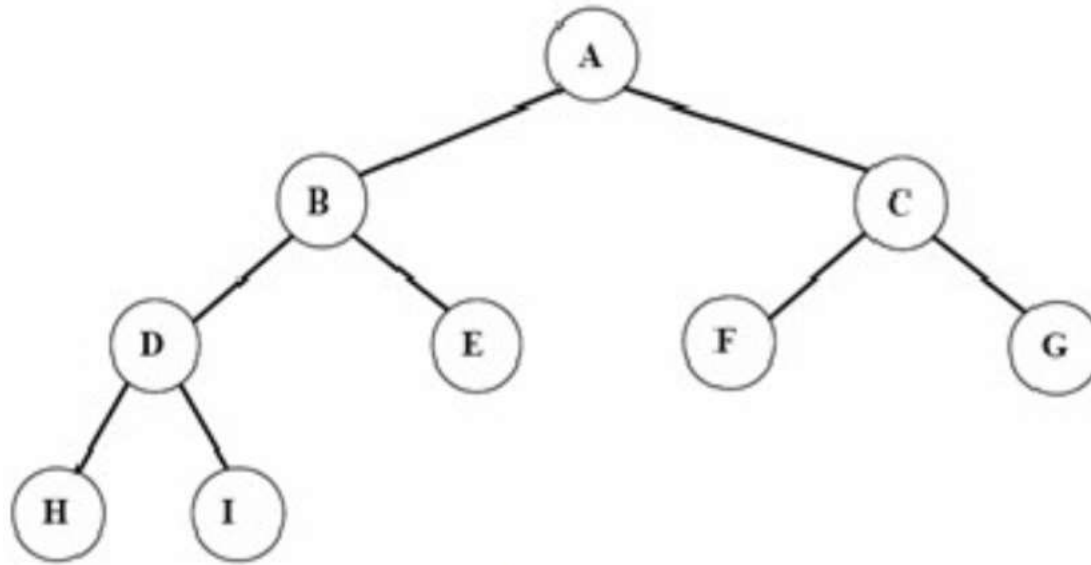


fig Binary tree

The preorder traversal output of the given tree is: A B D H I E C F G

The preorder is also known as depth first order.

In-order traversal

- The in-order traversal of a nonempty binary tree is defined as follows:
 - Traverse the left sub-tree in in-order
 - Visit the root node
 - Traverse the right sub-tree in inorder

- The in-order traversal output of the given tree is
H D I B E A F C G

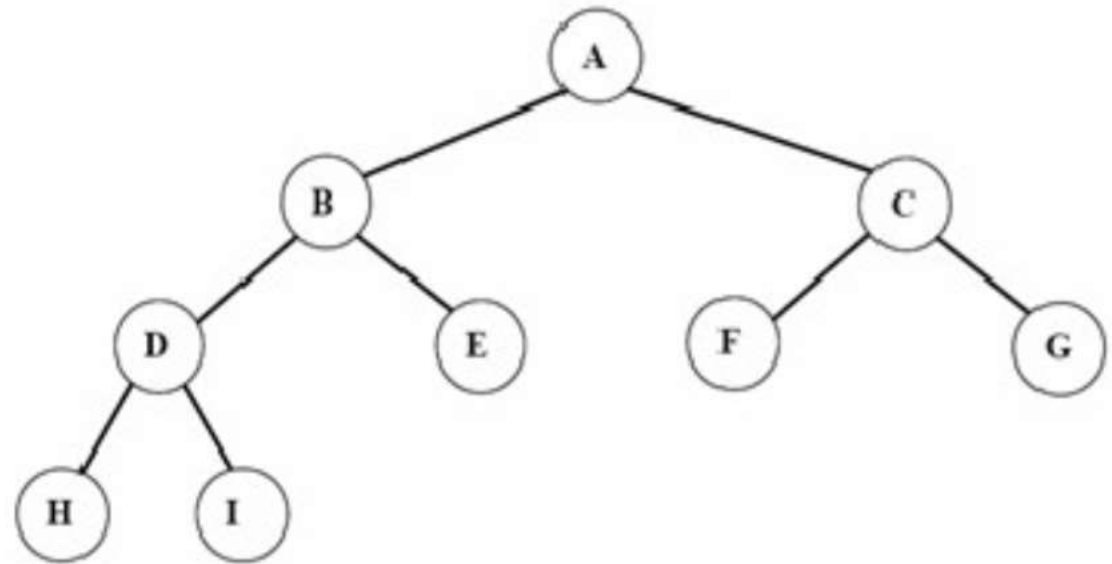


fig Binary tree

Post-order traversal

- The in-order traversal of a nonempty binary tree is defined as follows:
 - Traverse the left sub-tree in post-order
 - Traverse the right sub-tree in post-order
 - Visit the root node

- The in-order traversal output of the given tree is
H I D E B F G C A

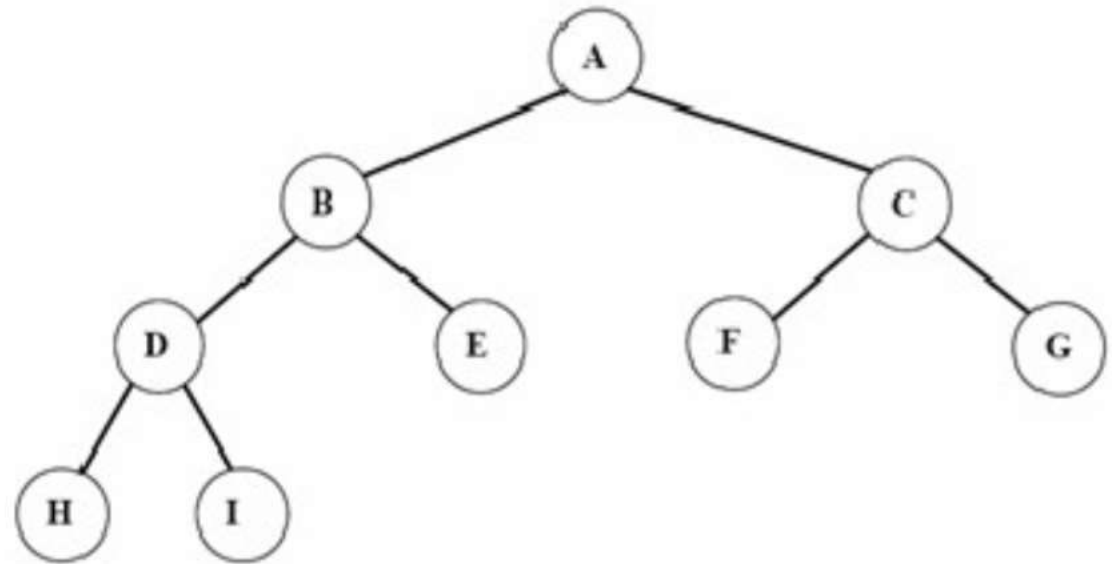
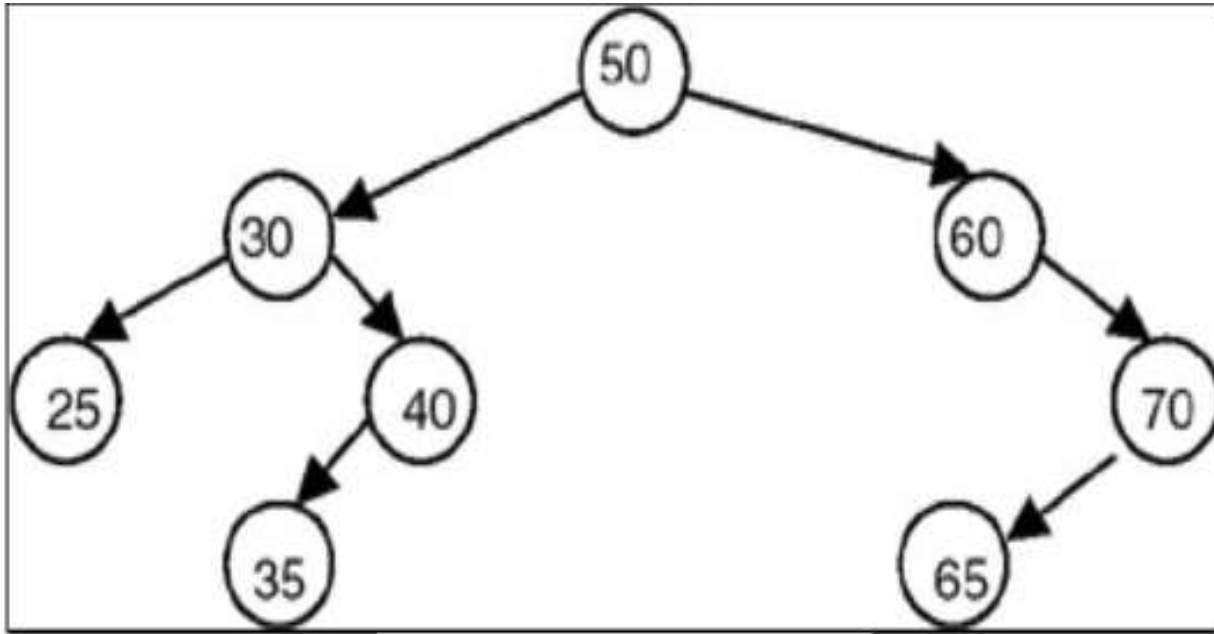


fig Binary tree

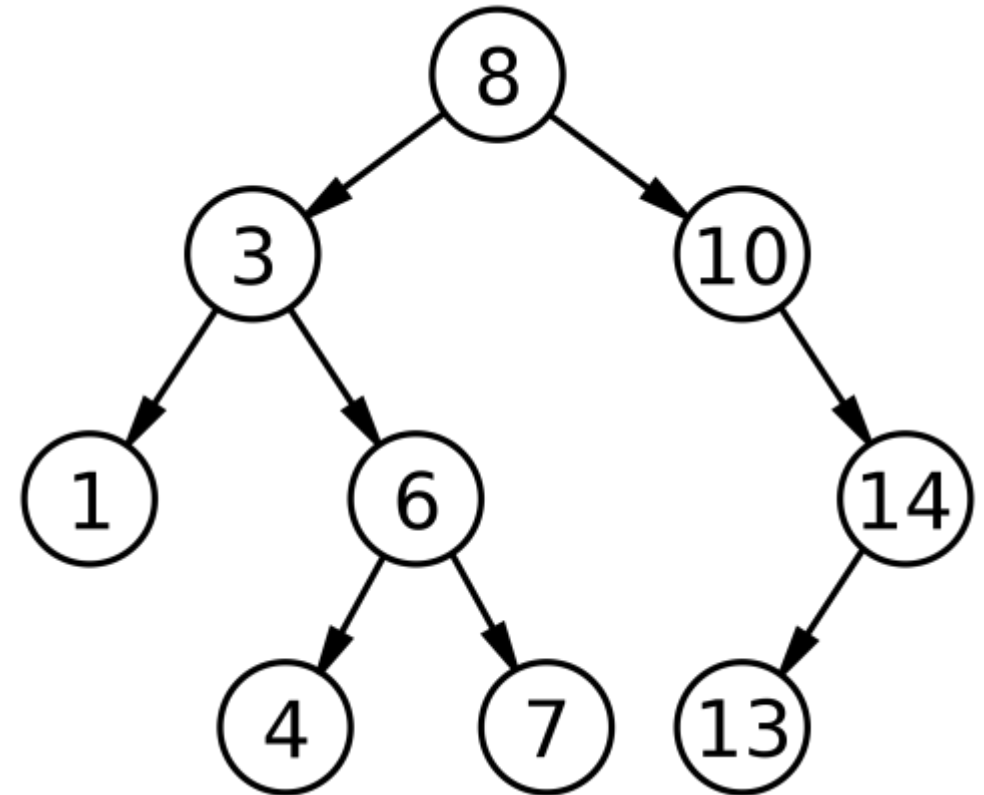
Binary Search Tree(BST)

- A binary search tree (BST) is a binary tree that is either empty or in which every node contains a key (value) and satisfies the following conditions:
 - All keys in the left sub-tree of the root are smaller than the key in the root node
 - All keys in the right sub-tree of the root are greater than the key in the root node
 - The left and right sub-trees of the root are again binary search trees

Binary Search Tree(BST)



The binary search tree.



Binary Search Tree(BST)

- A binary search tree is basically a binary tree, and therefore it can be traversed in inorder, preorder and postorder.
- If we traverse a binary search tree in inorder and print the identifiers contained in the nodes of the tree, we get a sorted list of identifiers in ascending order.

Why Binary Search Tree?

- Let us consider a problem of searching a list.
- If a list is ordered searching becomes faster if we use contiguous list(array).
- But if we need to make changes in the list, such as inserting new entries or deleting old entries, (**SLOWER!!!!**) because insertion and deletion in a contiguous list requires moving many of the entries every time.

Why Binary Search Tree?

- So we may think of using a linked list because it permits insertion and deletion to be carried out by adjusting only few pointers.
- But in an n-linked list, there is no way to move through the list other than one node at a time, permitting only sequential access.
- Binary trees provide an excellent solution to this problem. By making the entries of an ordered list into the nodes of a binary search tree, we find that we can search for a key in $O(\log n)$

Operations on Binary Search Tree (BST)

- Following operations can be done in BST:
 - **Search(k, T)**: Search for key k in the tree T. If k is found in some node of tree then return true otherwise return false.
 - **Insert(k, T)**: Insert a new node with value k in the info field in the tree T such that the property of BST is maintained.
 - **Delete(k, T)**: Delete a node with value k in the info field from the tree T such that the property of BST is maintained.
 - **FindMin(T), FindMax(T)**: Find minimum and maximum element from the given nonempty BST.

Searching Through The BST

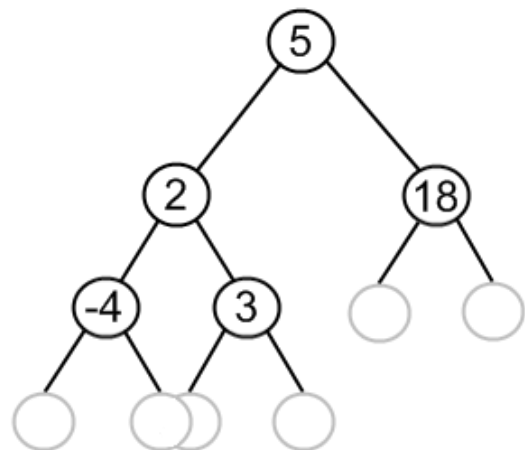
- Compare the target value with the element in the root node
 - ✓ If the target value is **equal**, the search is successful.
 - ✓ If target value is **less**, search the left subtree.
 - ✓ If target value is **greater**, search the right subtree.
 - ✓ If the subtree is **empty**, the search is unsuccessful.

Insertion of a node in BST

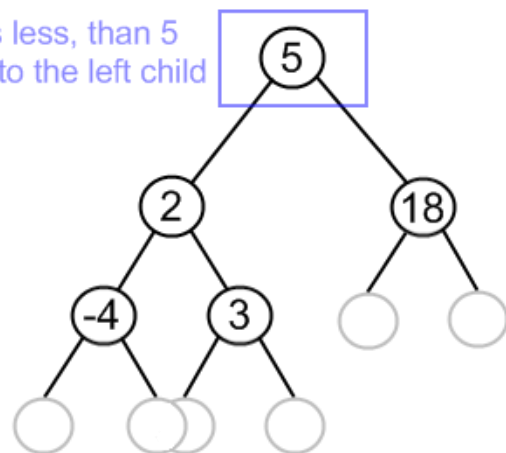
- To insert a new item in a tree, we must first verify that its key is different from those of existing elements.
- If a new value is less, than the current node's value, go to the left subtree, else go to the right subtree.
- Following this simple rule, the algorithm reaches a node, which has no left or right subtree.
- By the moment a place for insertion is found, we can say for sure, that a new value has no duplicate in the tree.

Algorithm for insertion in BST

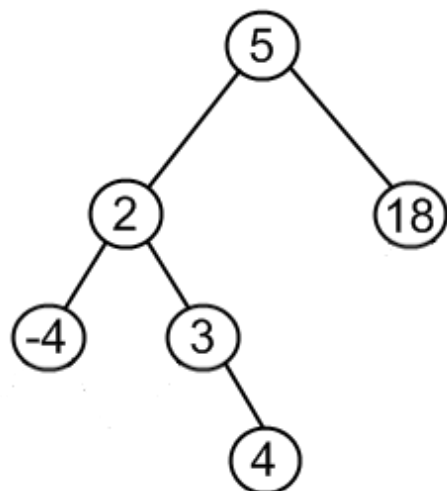
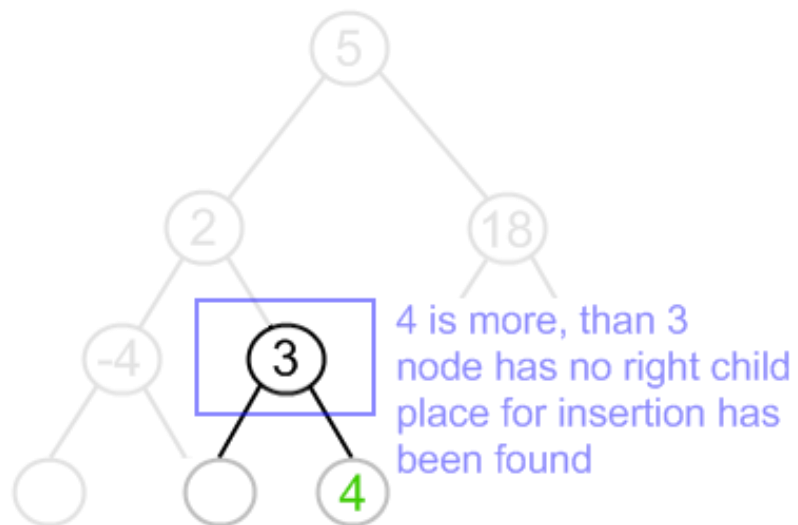
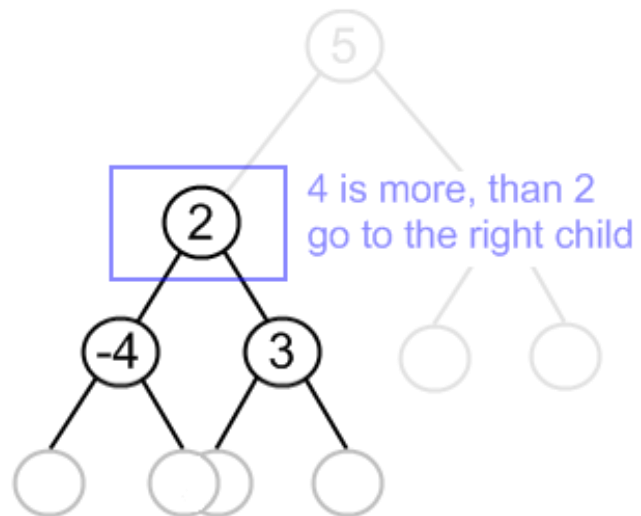
- Check, whether value in current node and a new value are equal. If so, duplicate is found. Otherwise,
- if a new value is less, than the node's value:
 - if a current node has no left child, place for insertion has been found;
 - otherwise, handle the left child with the same algorithm.
- if a new value is greater, than the node's value:
 - if a current node has no right child, place for insertion has been found;
 - otherwise, handle the right child with the same algorithm.



4 is less, than 5
go to the left child

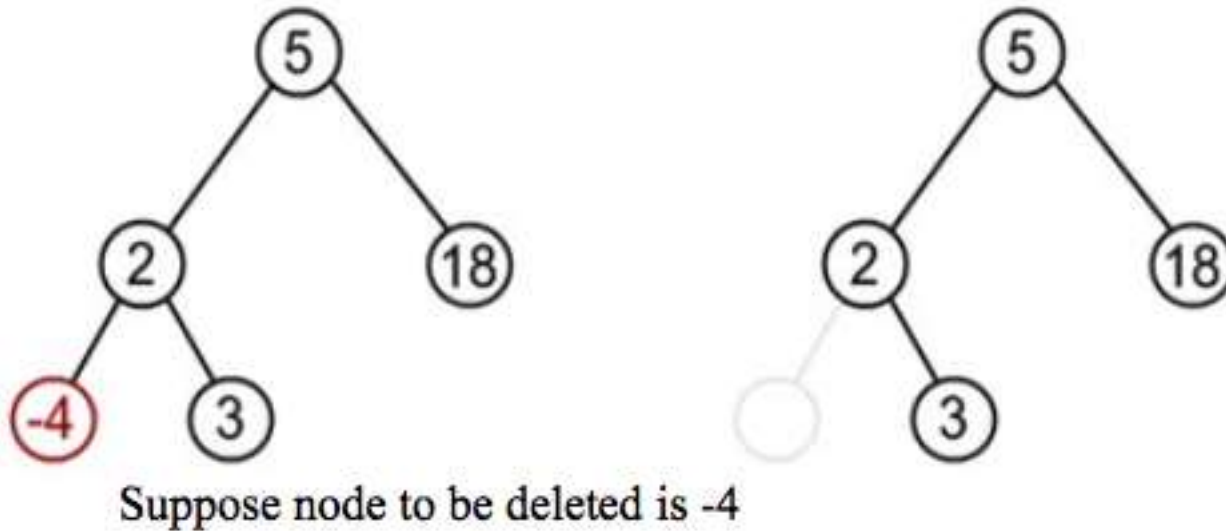


4 is more, than 2
go to the right child



Deleting a node from the BST

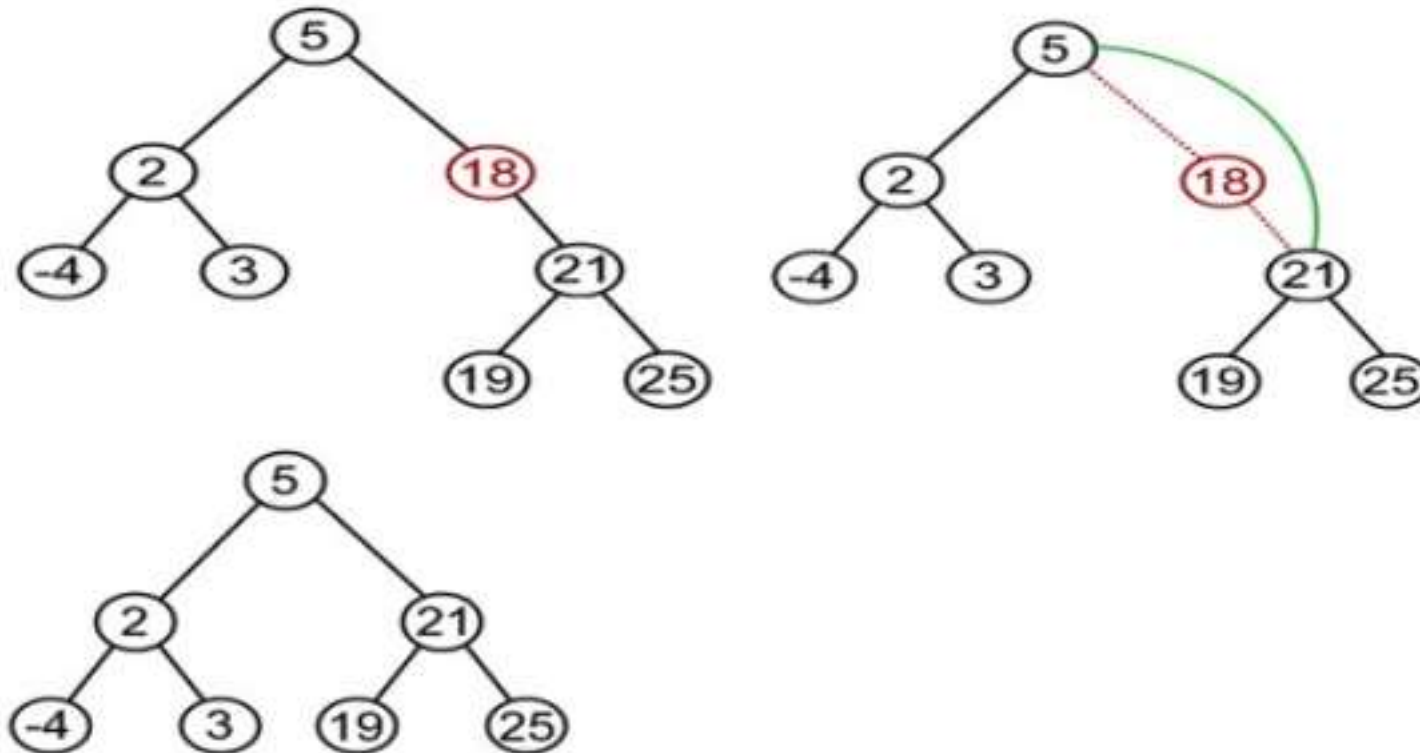
- While deleting a node from BST, there may be three cases:
 1. The node to be deleted may be a leaf node:
 - In this case simply delete a node and set null pointer to its parents those side at which this deleted node exist.



Deleting a node from the BST

2. The node to be deleted has one child

- In this case the child of the node to be deleted is appended to its parent node.
Suppose node to be deleted is 18



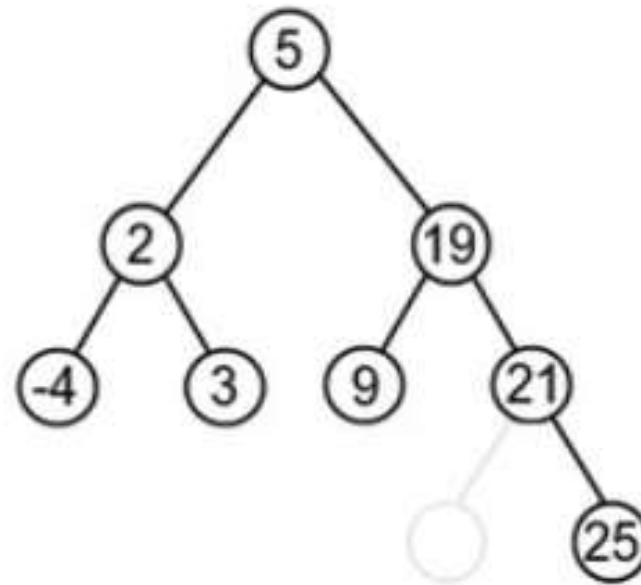
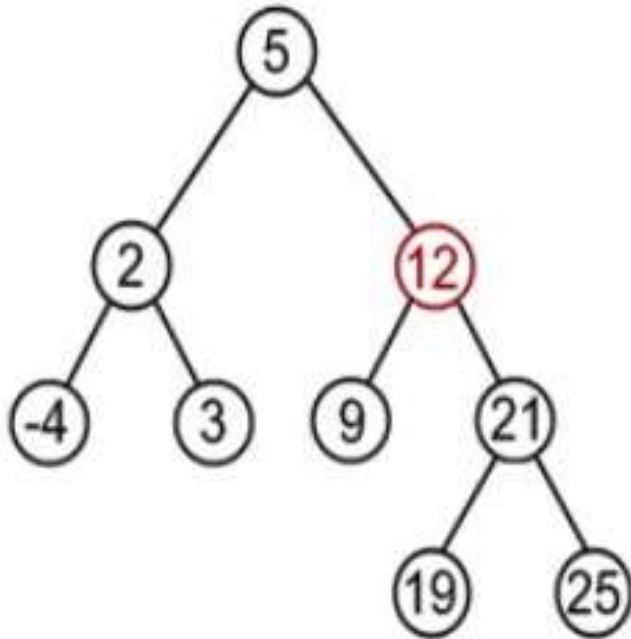
Deleting a node from the BST

3. the node to be deleted has two children:

In this case node to be deleted is replaced by its in-order successor node.

OR

If the node to be deleted is either replaced by its right sub-tree's leftmost node or its left sub-tree's rightmost node.



Suppose node to be deleted is 12

Find minimum element in the right sub-tree of the node to be removed. In current example it is 19.

General algorithm to delete a node from a BST:

1. start
2. if a node to be deleted is a leaf node at left side then simply delete and set null pointer to its parent's left pointer.
3. If a node to be deleted is a leaf node at right side then simply delete and set null pointer to its parent's right pointer
4. if a node to be deleted has one child then connect its child pointer with its parent pointer and delete it from the tree
5. if a node to be deleted has two children then replace the node being deleted either by
 - a. right most node of its left sub-tree or
 - b. left most node of its right sub-tree.
6. End